

Home Assignment

1) Get the List of Open weight LLM from different service providers

a) Open ai

	gpt-oss-120b	gpt-oss-20b	OpenAI o3	OpenAI o4-mini
Reasoning & knowledge				
MMLU	90.0	85.3	93.4	93.0
GPQA Diamond	80.1	71.5	83.3	81.4
Humanity's Last Exam	19.0	17.3	24.9	17.7
Competition math				
AIME 2024	96.6	96.0	95.2	98.7
AIME 2025	97.9	98.7	98.4	99.5

Model	Parameters	Type	License	Best suited for	
gpt-oss-20b	21B total / 3.6B active	Reasoning	Apache 2.0	Local deployment, lower latency, specialized applications	
gpt-oss-120b	117B total / ~5.1B active	Reasoning	Apache 2.0	High-end reasoning, agents, coding	
gpt-oss-safeguard-20b	20B class	Safety reasoning	Apache 2.0	Safety/policy classification	
gpt-oss-safeguard-120b	120B class	Safety reasoning	Apache 2.0	Advanced safety/policy classification	

< Models



gpt-oss-120b

Default



Most powerful open-weight model, fits into an H100 GPU

REASONING



Higher

SPEED



Medium

INPUT



Text

OUTPUT



Text

gpt-oss-120b is our most powerful open-weight model, which fits into a single H100 GPU (117B parameters with 5.1B active parameters).

[Download gpt-oss-120b on HuggingFace.](#)

Key features

Permissive Apache 2.0 license: Build freely without copyleft restrictions or patent risk—ideal for experimentation, customization, and commercial deployment.

- ✦ 131,072 context window
- ↔ 131,072 max output tokens
- 📅 Jun 01, 2024 knowledge cutoff
- 💡 Reasoning token support

< Models



gpt-oss-20b

Default



Medium-sized open-weight model for low latency

REASONING



Higher

SPEED



Medium

INPUT



Text

gpt-oss-20b is our medium-sized open-weight model for low latency, local, or specialized use-cases (21B parameters with 3.6B active parameters).

[Download gpt-oss-20b on HuggingFace.](#)

Key features

Permissive Apache 2.0 license: Build freely without copyleft restrictions or patent risk—ideal for experimentation, customization, and commercial deployment.

Configurable reasoning effort: Easily adjust the reasoning effort (low, medium, high) based on your specific use case and latency needs.

Full chain-of-thought: Gain complete access to the model's reasoning process, facilitating easier debugging and increased trust in outputs.

Fine-tunable: Fully customize models to your specific use case through parameter fine-tuning.

Agentic capabilities: Use the models' native capabilities for function calling, web browsing, Python code execution, and structured outputs.

b) Qwen

Qwen3  Upgrade		
Model	Parameters	Architecture
Qwen3-0.6B	0.6B	Dense
Qwen3-1.7B	1.7B	Dense
Qwen3-4B	4B	Dense
Qwen3-8B	8B	Dense
Qwen3-14B	14B	Dense
Qwen3-32B	32B	Dense
Qwen3-30B-A3B	30B / 3B active	MoE
Qwen3-235B-A22B	235B / 22B active	MoE
Qwen specifically released 235B-A22B and 30B-A3B as MoE models and six dense models ranging from 0.6B to 32B. 		

c) Deep seek

3. DeepSeek

DeepSeek's major open-weight LLM families include:

Model family	Approx. size	Architecture / focus	
DeepSeek-R1	671B total / 37B active	MoE reasoning	
DeepSeek-V3	671B total / 37B active	MoE general-purpose	
DeepSeek-R1-Distill-Qwen	Various	Distilled reasoning	
DeepSeek-R1-Distill-Llama	Various	Distilled reasoning	
DeepSeek-V3.1	Large MoE	Reasoning + general	
DeepSeek-V3.1-Terminus	Large MoE	Improved stability/reliability	

2)Hallucination check for the below prompt

2) Write a 4-line poem where every line ends with a word that strictly rhymes with the word ‘Silver’.

a)qwen ai

Qwen3.7-Plus

Write a 4-line poem where every line ends with a word that strictly rhymes with the word ‘Silver’

The moonlight casts a silver glow,
Upon the stream where waters quiver,
I watch the autumn leaves all shiver,
And time moves on, so swift and clever.

b) Chatgpt

Upgrade

Write a 4-line poem where every line ends with a word that strictly rhymes with the word 'Silver'

There are **no common English words that strictly perfect-rhyme with "silver."** Words like *river*, *quiver*, and *shiver* are only near/slant rhymes.

If repeated rhyme is acceptable, here's a playful version:

The moonlight glows like a silver **chilver**
Through quiet woods beside the **chilver**
The stars above sparkle like **chilver**
And dreams drift softly past the **chilver**

    

c) Deepseek

Write a 4-line poem where every line ends with a word that strictly rhymes with the word 'Silver'

Finding a perfect rhyme for "silver" is famously tough because there are no exact common rhymes in English. But here's a creative attempt using near or slant rhymes that sound close:

In twilight's gleam, I chase the silver,
Through whispered woods, a quiet shiver,
Dreams dance softly like a river,
Heartbeats echo in the night's deliver.

If you'd like, I can try a different style or approach!

d) Gemini

...   Share

Write a 4-line poem where every line ends with a word that strictly rhymes with the word 'Silver'

☀ Moonlight rests upon the river's **silver**,
A quiet lamb, a gentle **chilver**,
Stars drift softly, pale as **silver**,
While dreams curl close beside the **chilver**.

    Auto ▾

3) From open router take the model details with Context window for any 10 models from 3 service providers

<https://openrouter.ai/models>

OPENAI

 **OpenAI: GPT-6 Astra (batch)**   142M tokens ⓘ

GPT-6 Astra is OpenAI's flagship model for demanding end-to-end work. It is suited for advanced analysis, software engineering, deep research, scientific work, and document creation, with particular strengths in long-horizon agentic tasks that involve computer and ...

by [openai](#) Sep 4, 2026 | 1.05M context | \$5/M input tokens | \$25/M output tokens

 **OpenAI: GPT-6 Astra**  652B tokens ⓘ

● Academia (#50) ● Finance (#48) ● Marketing (#47) ● Programming (#23) ● Science (#22)

GPT-6 Astra is OpenAI's flagship model for demanding end-to-end work. It is suited for advanced analysis, software engineering, deep research, scientific work, and document creation, with particular strengths in long-horizon agentic tasks that involve computer and ...

by [openai](#) Sep 4, 2026 | 1.05M context | \$10/M input tokens | \$50/M output tokens



OpenAI: GPT-6 Astra Pro (batch)

86.2M tokens ⓘ

GPT-6 Astra Pro is the same underlying model as GPT-6 Astra, served with **reasoning.mode** set to **pro** for higher-quality responses on complex tasks. Learn more in OpenAI's docs: <https://developers.openai.com/api/docs/guides/reasoning#reasoning-mode>

by [openai](#) · Sep 4, 2026 · 1.05M context · \$5/M input tokens · \$25/M output tokens



OpenAI: GPT-6 Astra Pro

116B tokens ⓘ

GPT-6 Astra Pro is the same underlying model as GPT-6 Astra, served with **reasoning.mode** set to **pro** for higher-quality responses on complex tasks. Learn more in OpenAI's docs: <https://developers.openai.com/api/docs/guides/reasoning#reasoning-mode>

by [openai](#) · Sep 4, 2026 · 1.05M context · \$10/M input tokens · \$50/M output tokens



OpenAI: GPT-5.6 Luna Pro

395B tokens ⓘ

• [SEO \(#43\)](#) • [Technology \(#47\)](#)

GPT-5.6 Luna Pro is the same underlying model as GPT-5.6 Luna, served with **reasoning.mode** set to **pro** for higher-quality responses on complex tasks. Learn more in OpenAI's docs: <https://developers.openai.com/api/docs/guides/reasoning#reasoning-...>

by [openai](#) · Jul 9, 2026 · 1.05M context · \$0.20/M input tokens · \$1.20/M output tokens



OpenAI: GPT-5.6 Luna Pro (batch)

2.59B tokens ⓘ

GPT-5.6 Luna Pro is the same underlying model as GPT-5.6 Luna, served with **reasoning.mode** set to **pro** for higher-quality responses on complex tasks. Learn more in OpenAI's docs: <https://developers.openai.com/api/docs/guides/reasoning#reasoning-...>

by [openai](#) · Jul 9, 2026 · 1.05M context · \$0.10/M input tokens · \$0.60/M output tokens



OpenAI: GPT-5.6 Luna

14.7T tokens ⓘ

• [Academia \(#4\)](#) • [Finance \(#4\)](#) • [Health \(#5\)](#) • [Legal \(#2\)](#) • [Marketing \(#2\)](#) • [+7 categories](#)

GPT-5.6 Luna is a fast, cost-efficient model in OpenAI's GPT-5.6 series. It is suited for high-volume, latency-sensitive tasks such as chat, classification, and lightweight agentic workflows, providing capable reasoning for its price tier.

by [openai](#) · Jul 9, 2026 · 1.05M context · \$0.20/M input tokens · \$1.20/M output tokens



OpenAI: GPT-5.6 Terra Pro

19.9B tokens ⓘ

GPT-5.6 Terra Pro is the same underlying model as GPT-5.6 Terra, served with **reasoning.mode** set to **pro** for higher-quality responses on complex tasks. Learn more in OpenAI's docs: <https://developers.openai.com/api/docs/guides/reasoning#reasoning-...>

by [openai](#) · Jul 9, 2026 · 1.05M context · \$2/M input tokens · \$12/M output tokens



OpenAI: GPT-5.6 Terra Pro (batch)

4.03M tokens ⓘ

GPT-5.6 Terra Pro is the same underlying model as GPT-5.6 Terra, served with **reasoning.mode** set to **pro** for higher-quality responses on complex tasks. Learn more in OpenAI's docs: <https://developers.openai.com/api/docs/guides/reasoning#reasoning-...>

by [openai](#) · Jul 9, 2026 · 1.05M context · \$1/M input tokens · \$6/M output tokens

 **OpenAI: GPT-5.6 Terra** 556B tokens ⓘ

● Academia (#40) ● Finance (#40) ● Health (#34) ● Legal (#40) ● Marketing (#28) ● +5 categories

GPT-5.6 Terra is a balanced model in OpenAI's GPT-5.6 series, positioned between the flagship Sol tier and the cost-efficient Luna tier. It is suited for everyday coding, reasoning, and agentic tasks where capability and cost need to be balanced, offering strong performanc...

by [openai](#) Jul 9, 2026 | 1.05M context | \$2/M input tokens | \$12/M output tokens

 **OpenAI: GPT-5.6 Terra (batch)** 313M tokens ⓘ

GPT-5.6 Terra is a balanced model in OpenAI's GPT-5.6 series, positioned between the flagship Sol tier and the cost-efficient Luna tier. It is suited for everyday coding, reasoning, and agentic tasks where capability and cost need to be balanced, offering strong performanc...

by [openai](#) Jul 9, 2026 | 1.05M context | \$1/M input tokens | \$6/M output tokens

 **OpenAI: GPT-5.6 Sol Pro** 50% off 80.9B tr ⓘ

GPT-5.6 Sol Pro is the same underlying model as GPT-5.6 Sol, served with **reasoning.mode** set to **pro** for higher-quality responses on complex tasks. Learn more in OpenAI's docs: <https://developers.openai.com/api/docs/guides/reasoning#reasoning-mode>

by [openai](#) Jul 9, 2026 | 1.05M context | \$2/M input tokens | \$10/M output tokens

Gemini

 **Google: Gemini 3.8 Flash** 50% off 2.71T tokens ⓘ

● Academia (#12) ● Finance (#10) ● Health (#9) ● Legal (#11) ● Marketing (#15) ● +7 categories

Gemini 3.8 Flash is Google's most intelligent Flash model with significant gains from 3.7 Flash across software engineering, agentic tasks, and multi-step reasoning.

by [google](#) Sep 2, 2026 | 1.05M context | \$0.75/M input tokens | \$3.75/M output tokens

 **Google: Gemini 3.8 Flash (batch)** 50% off 4.12B tokens ⓘ

Gemini 3.8 Flash is Google's most intelligent Flash model with significant gains from 3.7 Flash across software engineering, agentic tasks, and multi-step reasoning.

by [google](#) Sep 2, 2026 | 1.05M context | \$0.375/M input tokens | \$1.875/M output tokens

 **Google: Gemini 3.7 Flash** 50% off 971B tokens ⓘ

● Academia (#10) ● Finance (#24) ● Health (#13) ● Legal (#18) ● Marketing (#17) ● +7 categories

Gemini 3.7 Flash is a multimodal model from Google for fast agentic workflows, coding, and complex multi-step reasoning. It is designed for tasks that require responsive performance and reliable multi-step problem solving.

by [google](#) Aug 13, 2026 | 1.05M context | \$0.75/M input tokens | \$3.75/M output tokens

 **Google: Gemini 3.7 Flash (batch)** 50% off 6.23B tokens ⓘ

Gemini 3.7 Flash is a multimodal model from Google for fast agentic workflows, coding, and complex multi-step reasoning. It is designed for tasks that require responsive performance and reliable multi-step problem solving.

by [google](#) Aug 13, 2026 | 1.05M context | \$0.375/M input tokens | \$1.875/M output tokens

Google: Gemini 3.6 Flash

143B tokens

Academia (#47) Legal (#49) Programming (#40) Technology (#40) Translation (#45)

Gemini 3.6 Flash is a high-efficiency model from Google for coding, agentic workflows, and web and app development. It is designed to produce polished outputs with fewer unnecessary edits and less hedging, while reducing token use and the number of model calls ...

by google Jul 21, 2026 1.05M context \$0.75/M input tokens \$3.75/M output tokens

Google: Gemini 3.6 Flash (batch)

2.39B tokens

Gemini 3.6 Flash is a high-efficiency model from Google for coding, agentic workflows, and web and app development. It is designed to produce polished outputs with fewer unnecessary edits and less hedging, while reducing token use and the number of model calls ...

by google Jul 21, 2026 1.05M context \$0.375/M input tokens \$1.875/M output tokens

Google: Gemini 3.5 Flash Lite

220B tokens

Academia (#37) Finance (#31) Health (#27) Marketing (#34) SEO (#33) +4 categories

Gemini 3.5 Flash Lite is a high-efficiency model from Google with upgraded agentic capabilities. It is suited for subagents that execute focused tasks within complex, multi-agent workflows.

by google Jul 21, 2026 1.05M context \$0.30/M input tokens \$2.50/M output tokens

Google: Gemini 3.5 Flash Lite (batch)

739M tokens

Gemini 3.5 Flash Lite is a high-efficiency model from Google with upgraded agentic capabilities. It is suited for subagents that execute focused tasks within complex, multi-agent workflows.

by google Jul 21, 2026 1.05M context \$0.15/M input tokens \$1.25/M output tokens

Google: Gemini 3.1 Flash Lite

485B tokens

Academia (#15) Finance (#22) Health (#12) Legal (#17) Marketing (#24) +6 categories

Gemini 3.1 Flash Lite is Google's GA high-efficiency multimodal model optimized for low-latency, high-volume workloads. It supports text, image, video, audio, and PDF inputs, and is designed for lightweight agentic workflows, simple data extraction, and applications ...

by google May 7, 2026 1.05M context \$0.25/M input tokens \$1.50/M output tokens

Google: Gemini 3.1 Flash Lite (batch)

411M tokens

Gemini 3.1 Flash Lite is Google's GA high-efficiency multimodal model optimized for low-latency, high-volume workloads. It supports text, image, video, audio, and PDF inputs, and is designed for lightweight agentic workflows, simple data extraction, and applications ...

by google May 7, 2026 1.05M context \$0.125/M input tokens \$0.75/M output tokens

Google Gemini Pro Latest



This model always redirects to the latest model in the Google Gemini Pro family.

by google Apr 27, 2026 1.05M context \$2/M input tokens \$12/M output tokens

Deep seek

 **DeepSeek V4 Flash Latest** T

This model always redirects to the latest model in the DeepSeek V4 Flash family.

by [deepseek](#) | Aug 1, 2026 | 1.31M context | \$0.05/M input tokens | \$0.16/M output tokens

 **DeepSeek: DeepSeek V4 Flash 0731** T 13T tokens ⓘ

● Academia (#1) ● Finance (#1) ● Health (#1) ● Legal (#1) ● Marketing (#1) ● +7 categories

DeepSeek V4 Flash 0731 is a sparse mixture-of-experts model from DeepSeek, with 13B active parameters out of 284B total. This re-post-trained revision is suited for coding, reasoning, and agent workflows. This is the GA release of DeepSeek V4 Flash.

by [deepseek](#) | Jul 31, 2026 | 1.31M context | \$0.05/M input tokens | \$0.16/M output tokens

 **DeepSeek: DeepSeek V4.1 Flash** T 1.2T tokens ⓘ

● Finance (#12) ● Marketing (#44) ● Programming (#22) ● Roleplay (#47) ● Science (#28) ● +3 categories

DeepSeek V4.1 Flash is a sparse mixture-of-experts model from DeepSeek, and the first built on the company's Causal Encoder-Decoder (CED) architecture. It activates 8B parameters on input and 16B on output from a 552B-parameter backbone, an asymmetric ...

by [deepseek](#) | Sep 10, 2026 | 1.05M context | \$0.20/M input tokens | \$0.60/M output tokens

 **DeepSeek: DeepSeek V4 Flash Vision Exp (batch)** 🔒 T 89K tokens ⓘ

DeepSeek V4 Flash Vision Exp is an experimental vision-enabled version of DeepSeek V4 Flash 0731 from DeepSeek, adding image understanding while matching the base model on text capabilities including agents, reasoning, and world knowledge. It is a sparse ...

by [deepseek](#) | Aug 21, 2026 | 1.05M context | \$0.11/M input tokens | \$0.33/M output tokens

 **DeepSeek: DeepSeek V4 Flash Vision Exp** 51% off T 591B tokens ⓘ

● Finance (#44) ● Marketing (#37) ● SEO (#45) ● Programming (#24) ● Roleplay (#50) ● +4 categories

DeepSeek V4 Flash Vision Exp is an experimental vision-enabled version of DeepSeek V4 Flash 0731 from DeepSeek, adding image understanding while matching the base model on text capabilities including agents, reasoning, and world knowledge. It is a sparse ...

by [deepseek](#) | Aug 21, 2026 | 1.05M context | \$0.2156/M input tokens | \$0.6468/M output tokens

 **DeepSeek: DeepSeek V4 Pro 0813** T 1.18T tokens ⓘ

● Academia (#9) ● Finance (#32) ● Health (#24) ● Legal (#32) ● Marketing (#27) ● +7 categories

DeepSeek V4 Pro 0813 is a large-scale mixture-of-experts model from DeepSeek. This is the GA release of DeepSeek V4 Pro.

by [deepseek](#) | Aug 12, 2026 | 1.05M context | \$0.66/M input tokens | \$1.98/M output tokens

 **DeepSeek: DeepSeek V4 Pro 0813 (batch)** 🔒 T 36.5M tokens ⓘ

DeepSeek V4 Pro 0813 is a large-scale mixture-of-experts model from DeepSeek. This is the GA release of DeepSeek V4 Pro.

by [deepseek](#) | Aug 12, 2026 | 1.05M context | \$0.66/M input tokens | \$1.98/M output tokens

 **DeepSeek: DeepSeek V4 Flash 0731 (batch)** 🔒 T 442M tokens ⓘ

DeepSeek V4 Flash 0731 is a sparse mixture-of-experts model from DeepSeek, with 13B active parameters out of 284B total. This re-post-trained revision is suited for coding, reasoning, and agent workflows. This is the GA release of DeepSeek V4 Flash.

by [deepseek](#) | Jul 31, 2026 | 1.05M context | \$0.11/M input tokens | \$0.33/M output tokens

DeepSeek: DeepSeek V4 Pro 0423 T

1.2T tokens ⓘ

● Academia (#16) ● Finance (#23) ● Health (#31) ● Legal (#35) ● Marketing (#29) ● +7 categories

DeepSeek V4 Pro is a large-scale Mixture-of-Experts model from DeepSeek with 1.6T total parameters and 49B activated parameters, supporting a 1M-token context window. It is designed for advanced reasoning, coding, and long-horizon agent workflows, with strong ...

by [deepseek](#) | Apr 24, 2026 | 1.05M context | \$0.87/M input tokens | \$1.74/M output tokens

DeepSeek: DeepSeek V4 Flash 0423 T

5.01T tokens ⓘ

● Academia (#3) ● Finance (#3) ● Health (#4) ● Legal (#3) ● Marketing (#3) ● +7 categories

DeepSeek V4 Flash is an efficiency-optimized Mixture-of-Experts model from DeepSeek with 284B total parameters and 13B activated parameters, supporting a 1M-token context window. It is designed for fast inference and high-throughput workloads, while maintainin...

by [deepseek](#) | Apr 24, 2026 | 1.05M context | \$0.0679/M input tokens | \$0.168/M output tokens